

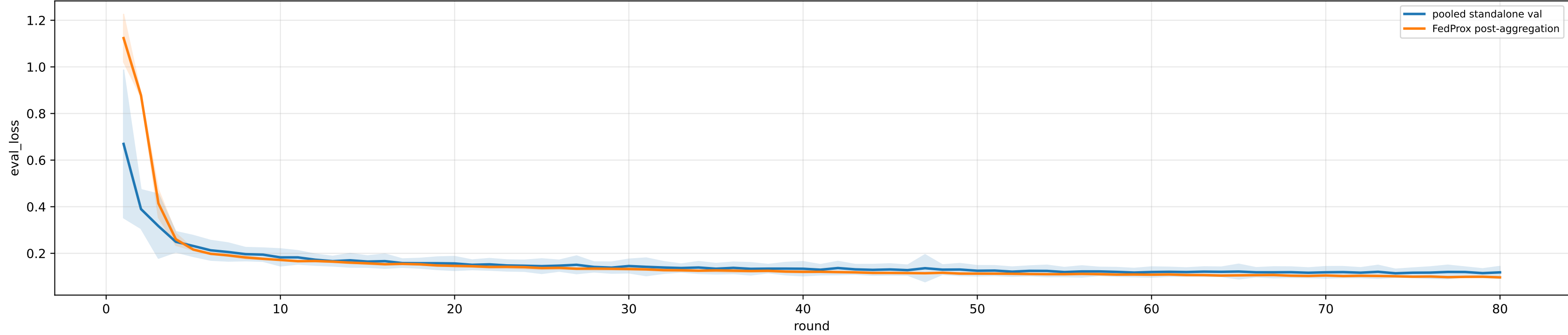
FedProx diagnostics | aggregated / pooled over 20 clients |
model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64

heterogeneity | task_jsd_mean=0.013 | task_jsd_median=0.009 | task_jsd_max=0.045 | family_centroid_jsd_mean=0.015
per-task pos-rate std | cycle2_std=0.100 | cycle3_std=0.079 | cycle4_std=0.074 | cycle5_std=0.084 | cycle6_std=0.095

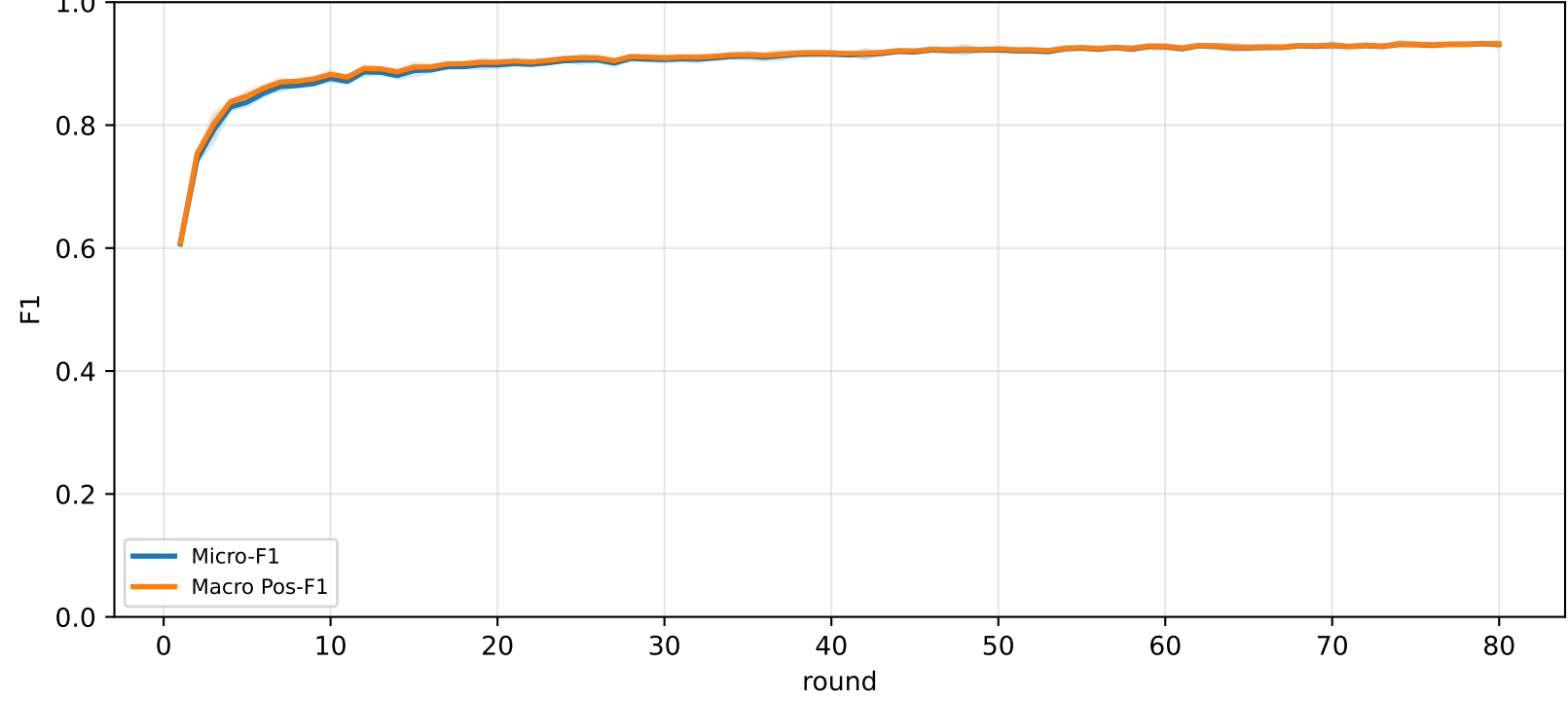
Best logged test performance (FedProx vs Local centralized)

Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	92.3% ± 0.2%	92.1% ± 0.2%	99.1% ± 0.0%	98.3% ± 0.1%	95.9% ± 0.2%	90.9% ± 0.2%	82.3% ± 0.8%
FedProx	93.2% ± 0.0%	92.8% ± 0.0%	99.3% ± 0.0%	98.5% ± 0.1%	95.7% ± 0.2%	91.4% ± 0.1%	81.3% ± 0.2%

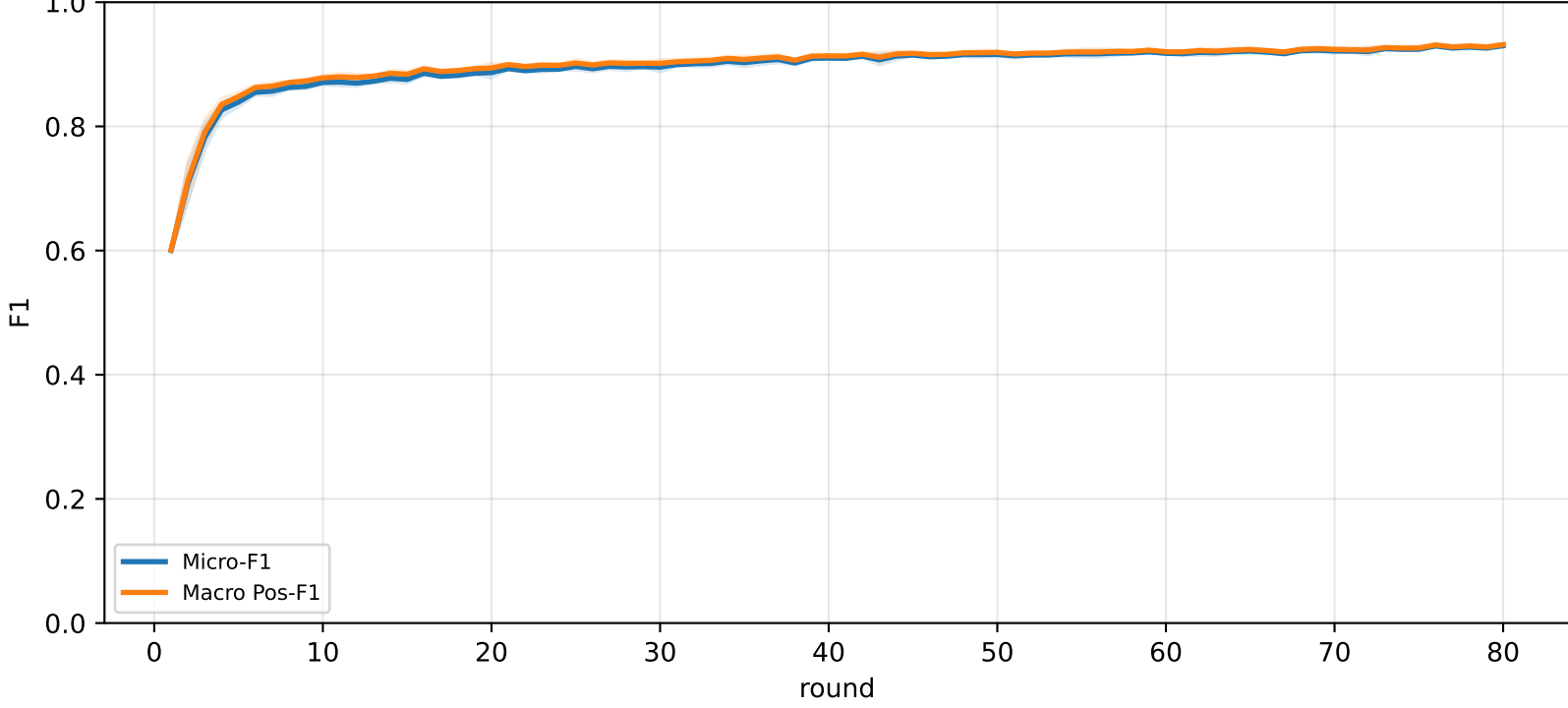
validation loss comparison



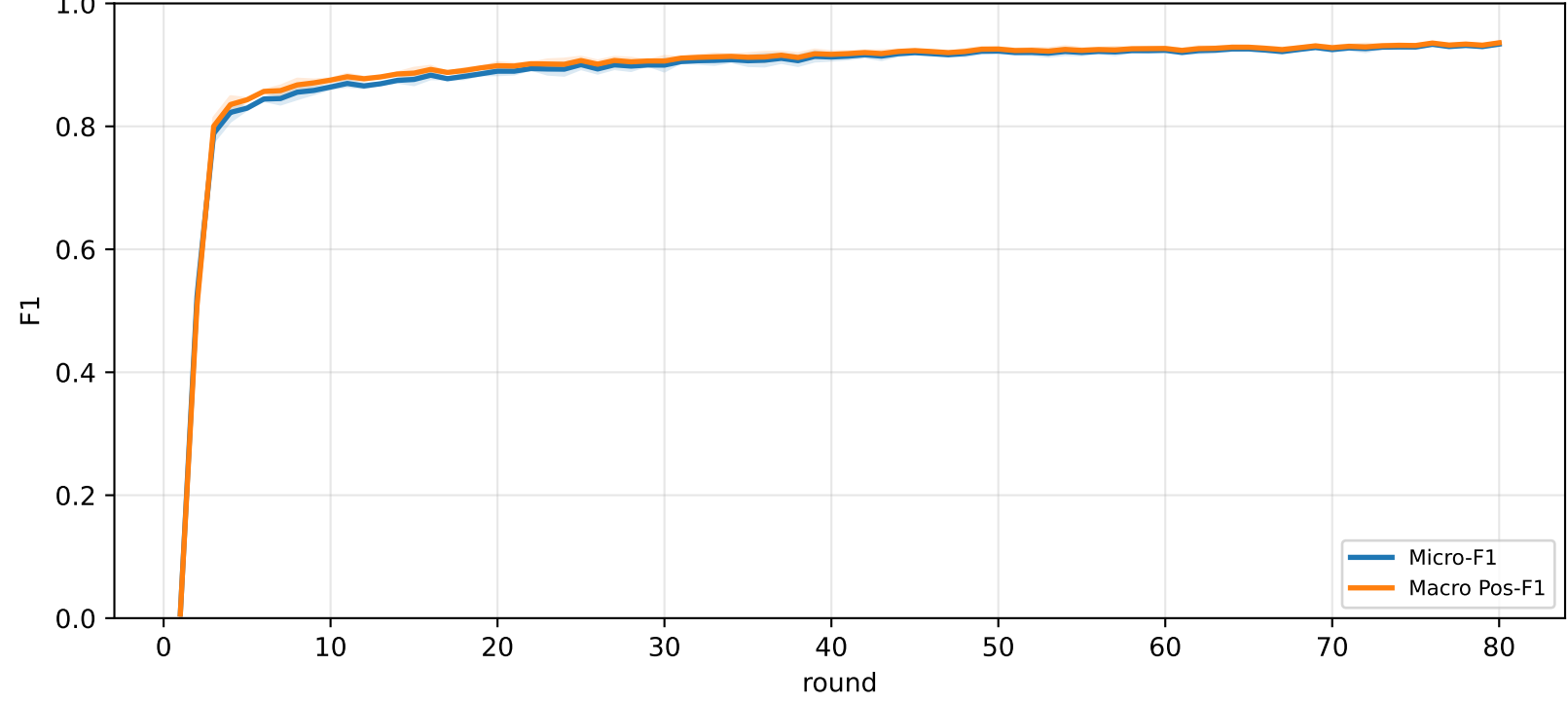
Pooled local centralized micro / macro F1



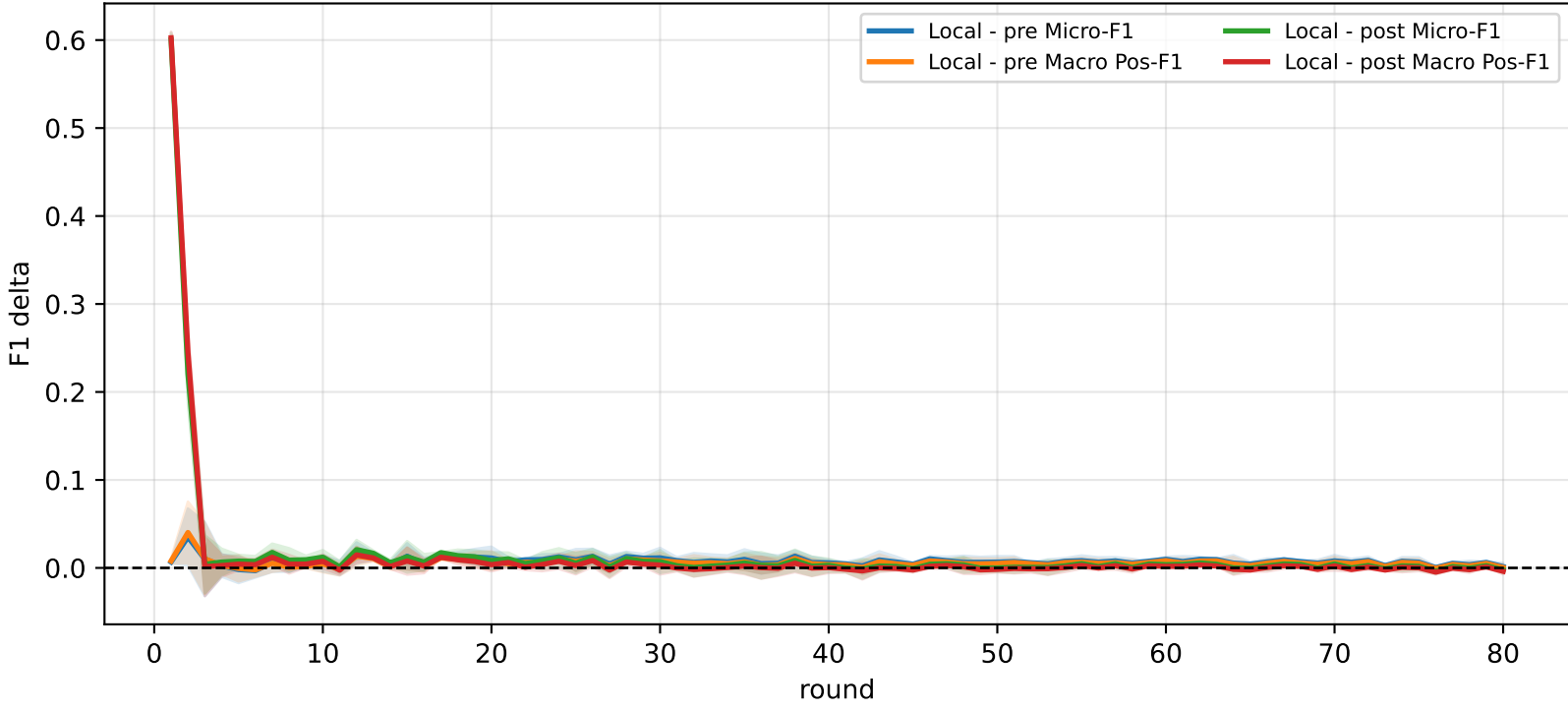
Pooled FedProx micro / macro F1 (pre-aggregation)



Pooled FedProx micro / macro F1 (post-aggregation)



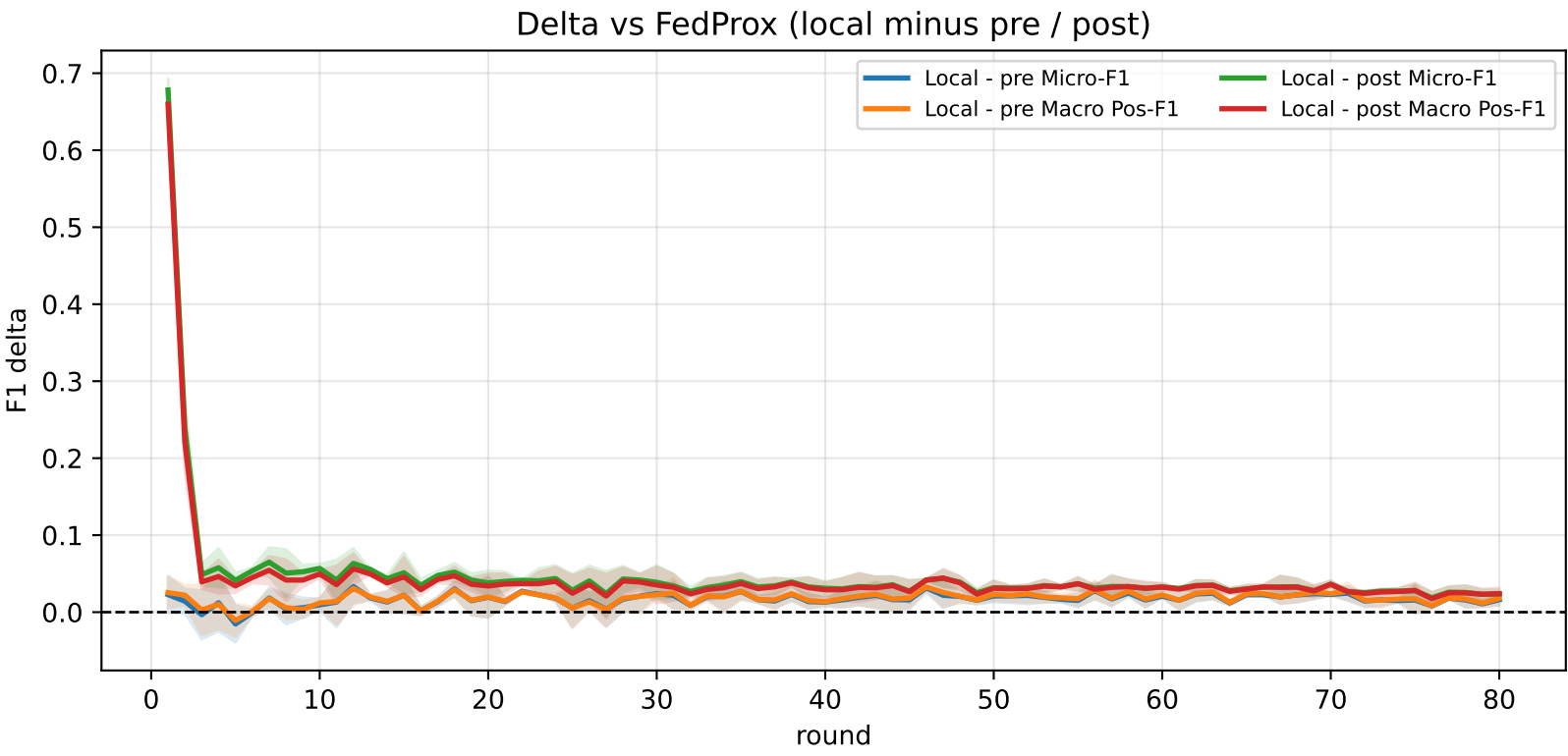
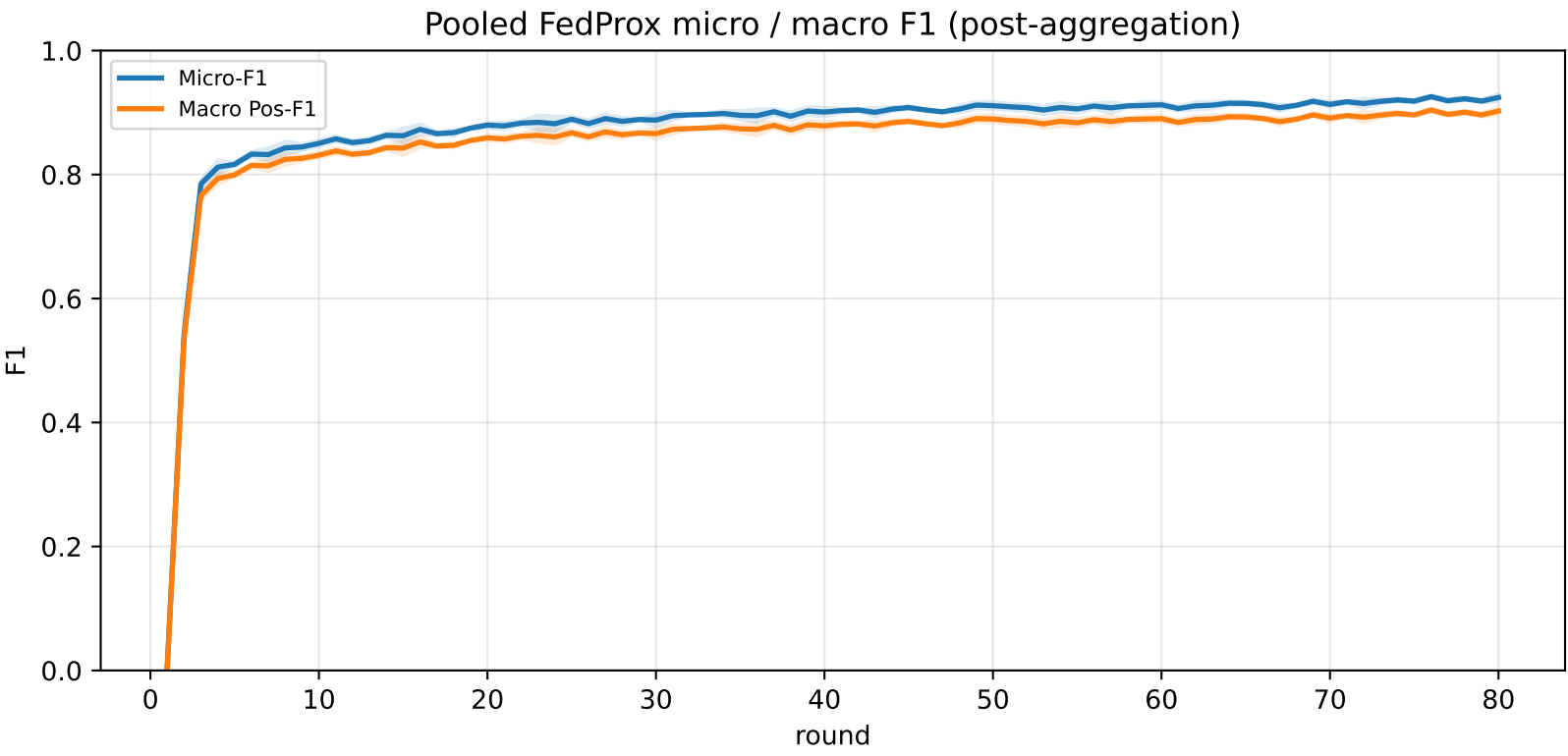
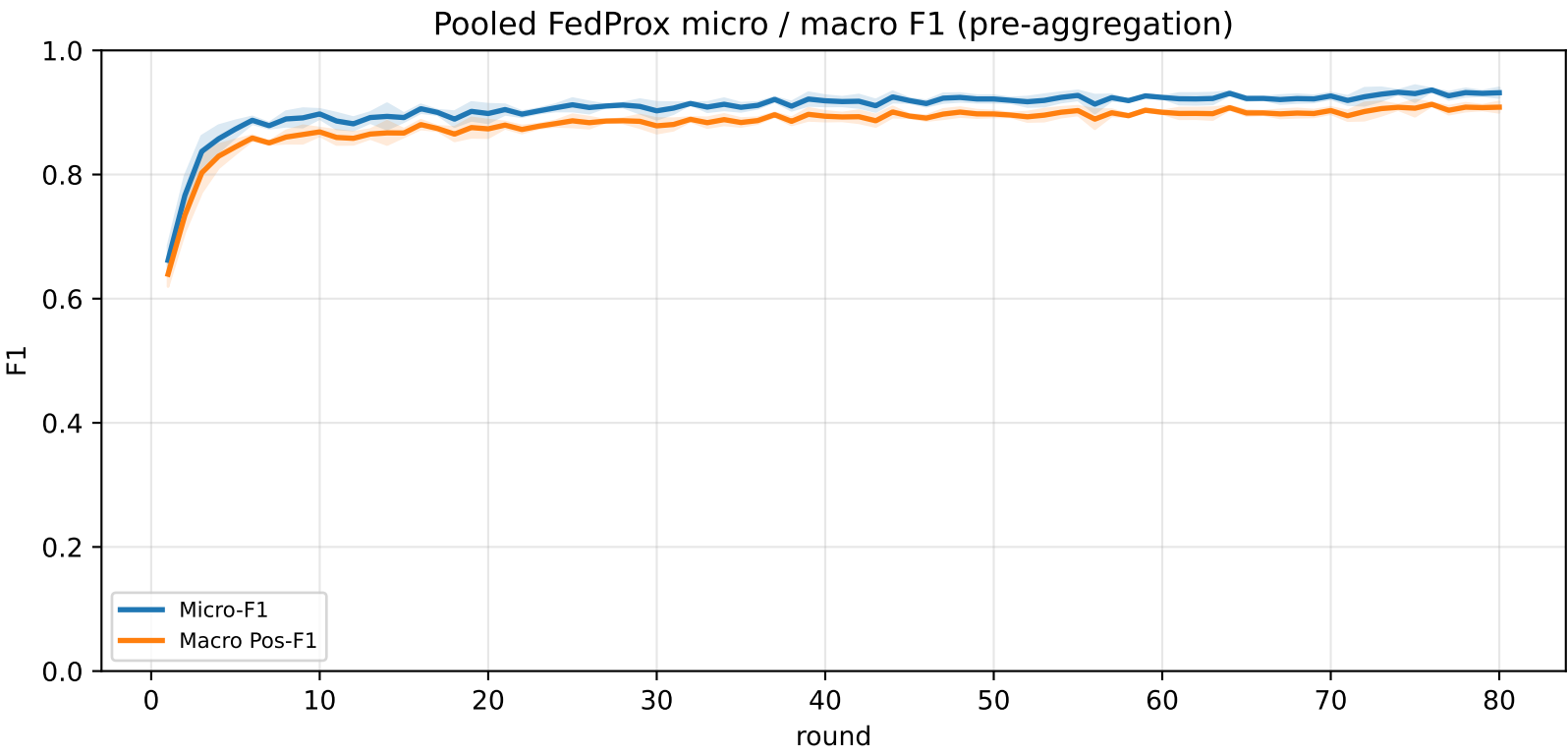
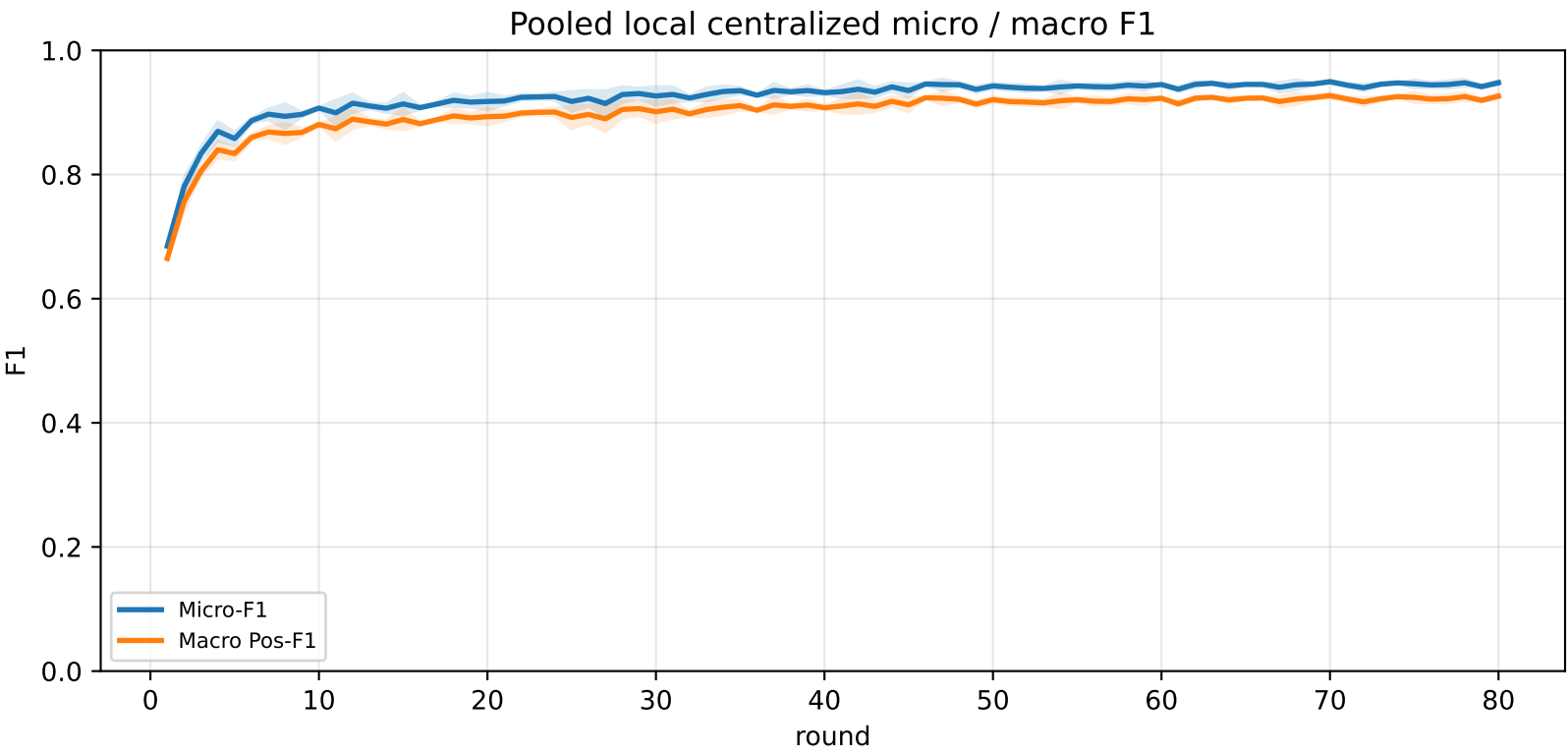
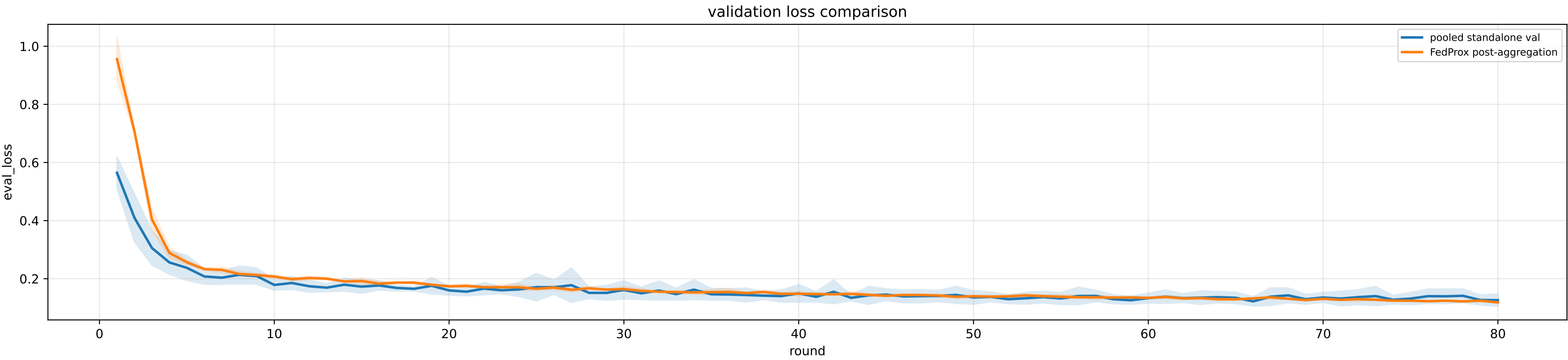
Delta vs FedProx (local minus pre / post)



FedProx diagnostics | family=strong_decreasing | n_clients=4 |
model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64

family=strong_decreasing | cycle2=37.8%±0.4% | cycle3=32.8%±0.8% | cycle4=27.9%±0.5% | cycle5=22.4%±0.5% | cycle6=16.0%±0.2%

Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	94.5% ± 0.2%	92.5% ± 0.3%	99.6% ± 0.0%	99.1% ± 0.2%	96.8% ± 0.3%	90.2% ± 0.2%	76.6% ± 1.6%
FedProx	91.9% ± 0.1%	89.9% ± 0.1%	99.7% ± 0.1%	99.1% ± 0.1%	96.4% ± 0.2%	87.7% ± 0.1%	66.6% ± 0.1%

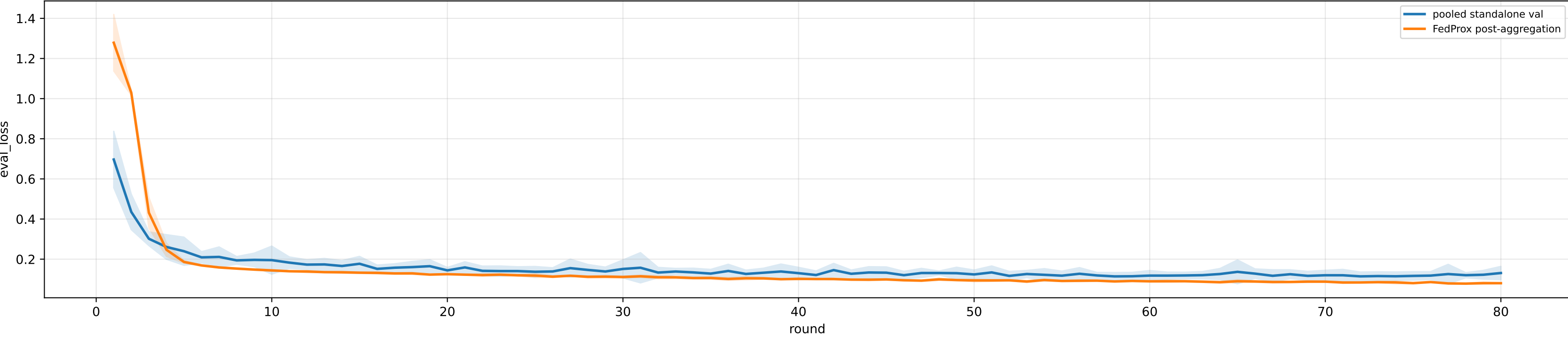


FedProx diagnostics | family=mild_decreasing | n_clients=4 | model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64

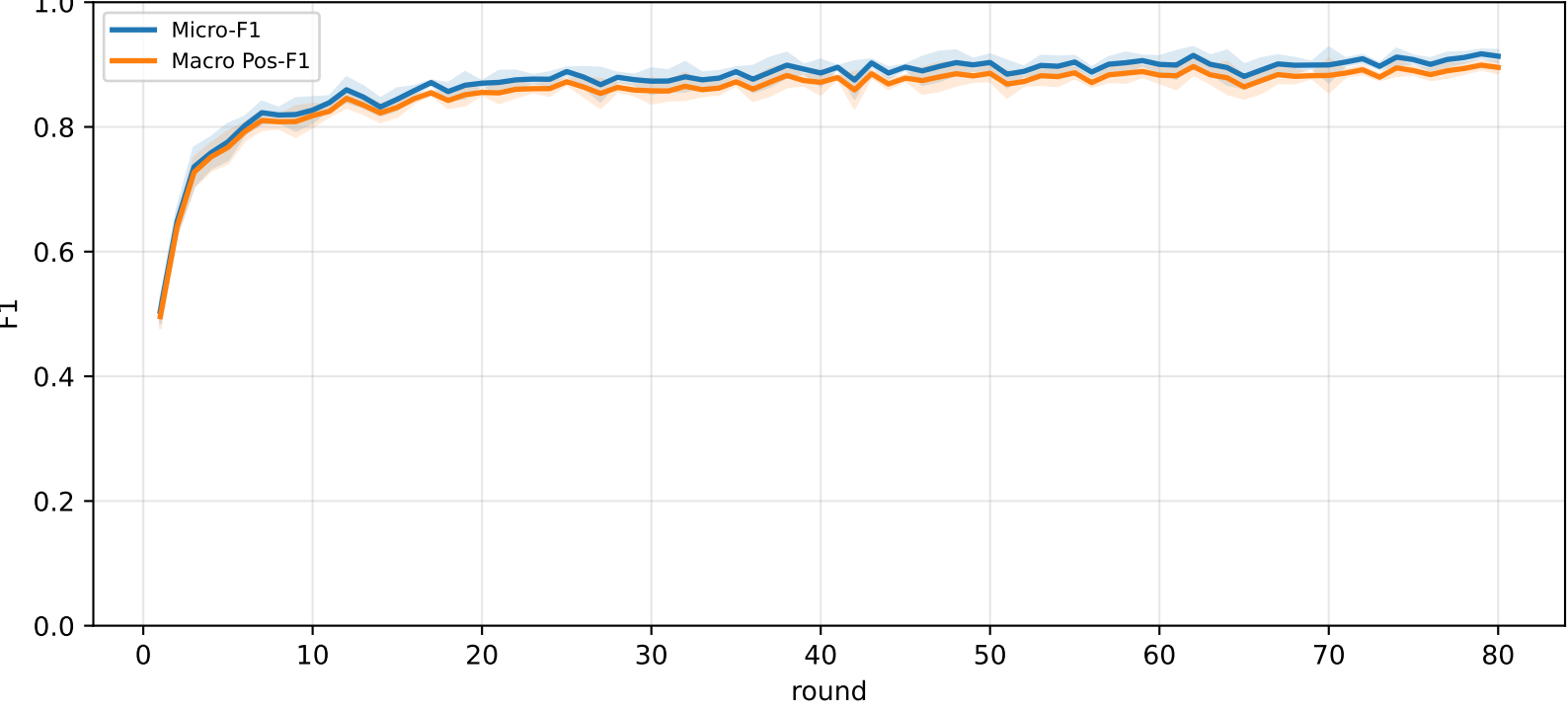
family=mild_decreasing | cycle2=17.1%±0.7% | cycle3=14.6%±1.1% | cycle4=12.6%±0.5% | cycle5=9.5%±0.8% | cycle6=7.7%±0.4%

Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	91.1% ± 1.1%	89.3% ± 1.1%	99.2% ± 0.1%	98.2% ± 0.1%	94.8% ± 0.6%	85.2% ± 2.2%	69.5% ± 4.1%
FedProx	93.9% ± 0.1%	92.3% ± 0.1%	99.5% ± 0.1%	99.0% ± 0.2%	96.8% ± 0.3%	91.8% ± 0.0%	75.0% ± 0.3%

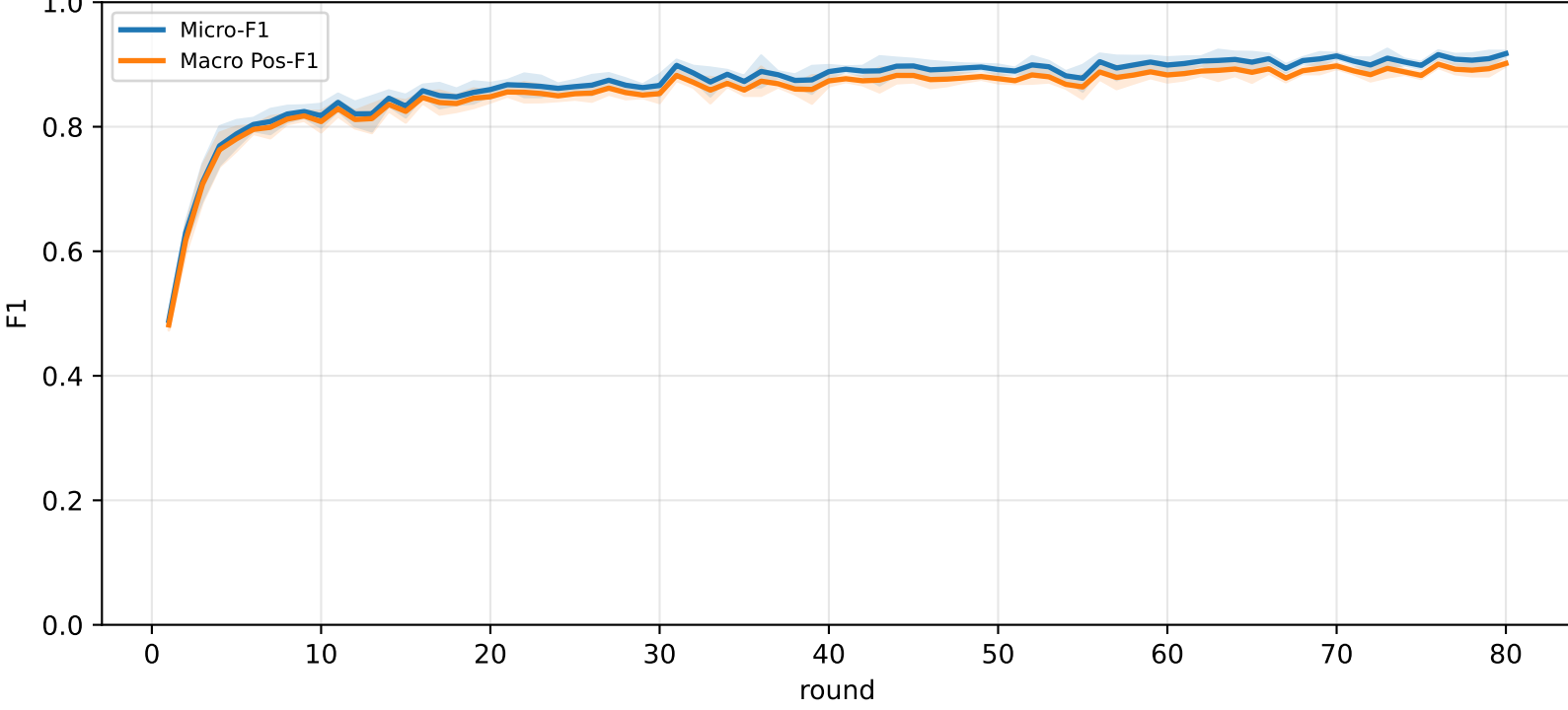
validation loss comparison



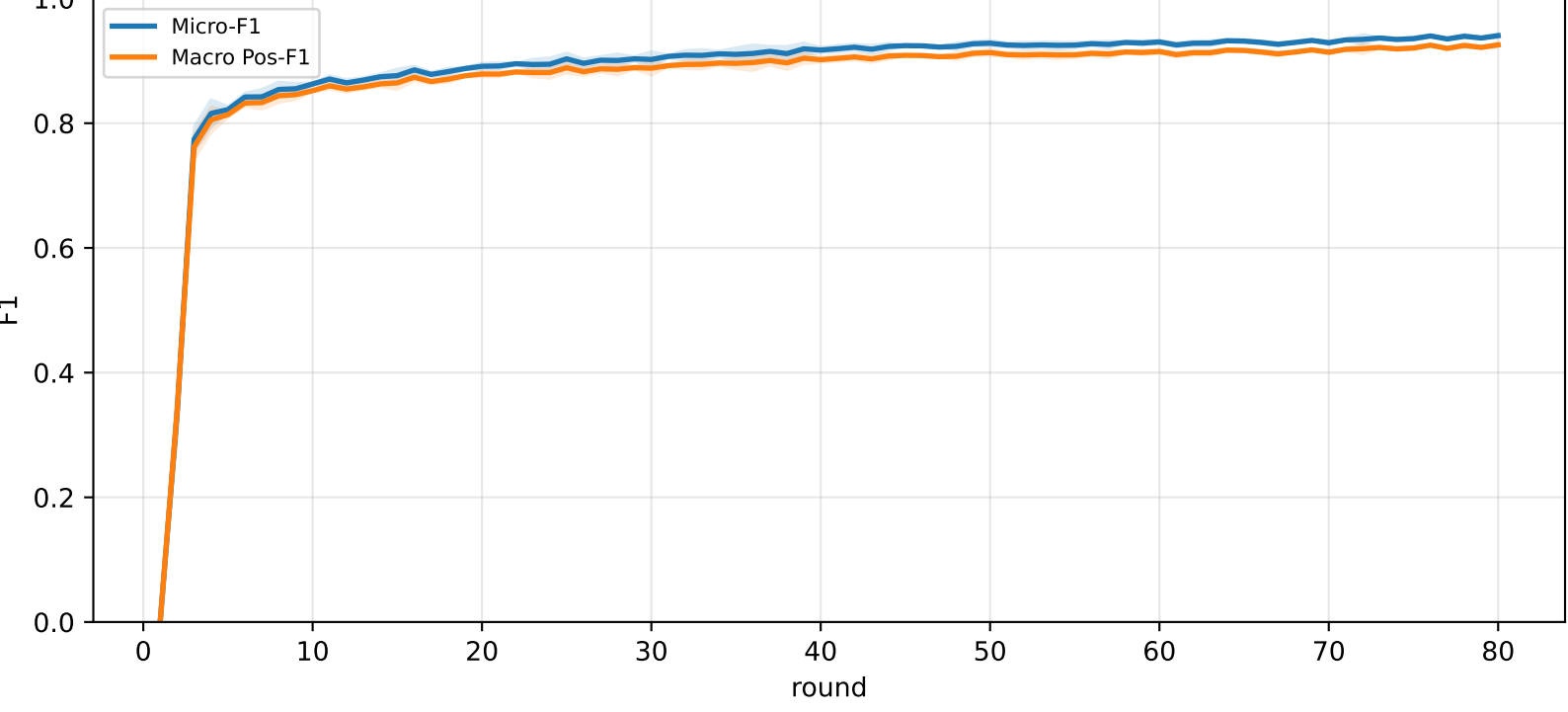
Pooled local centralized micro / macro F1



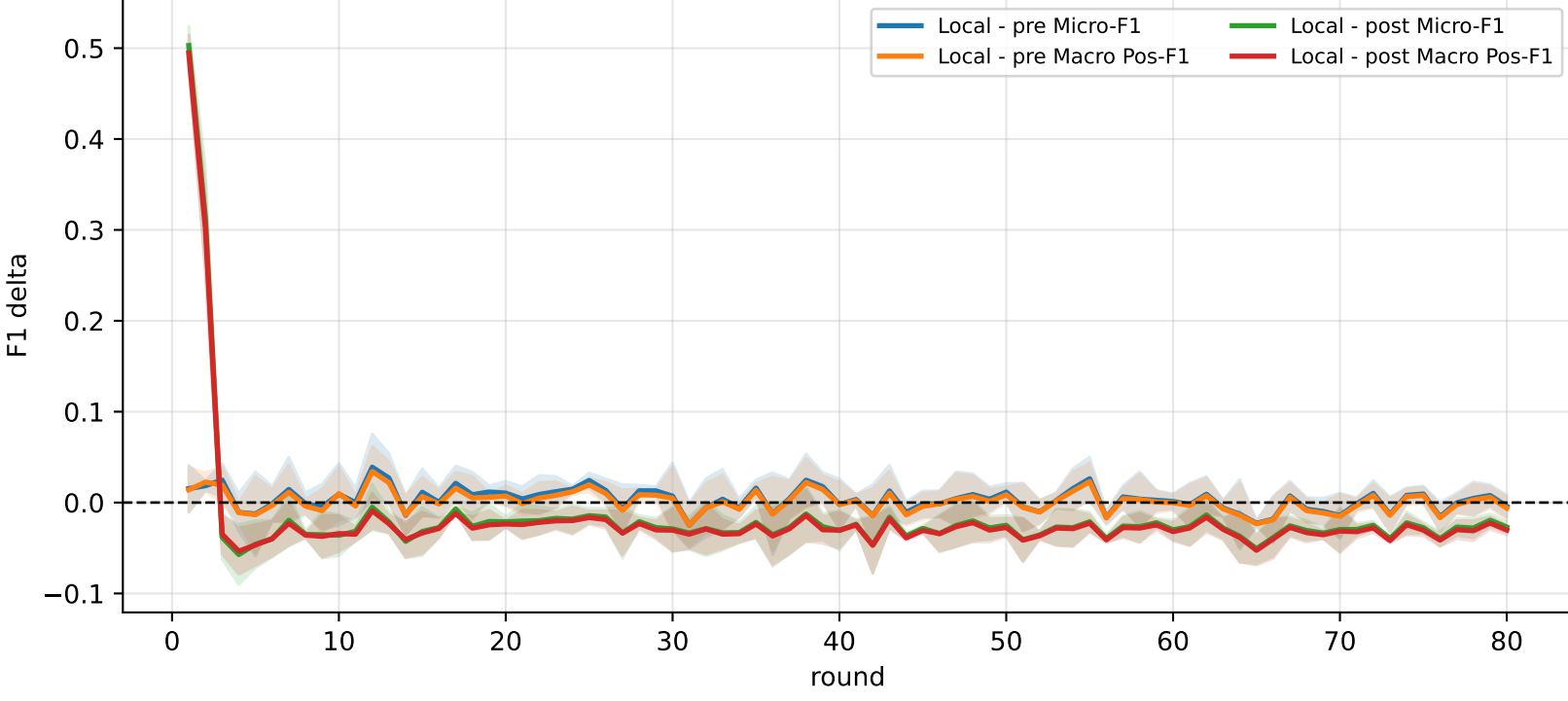
Pooled FedProx micro / macro F1 (pre-aggregation)



Pooled FedProx micro / macro F1 (post-aggregation)



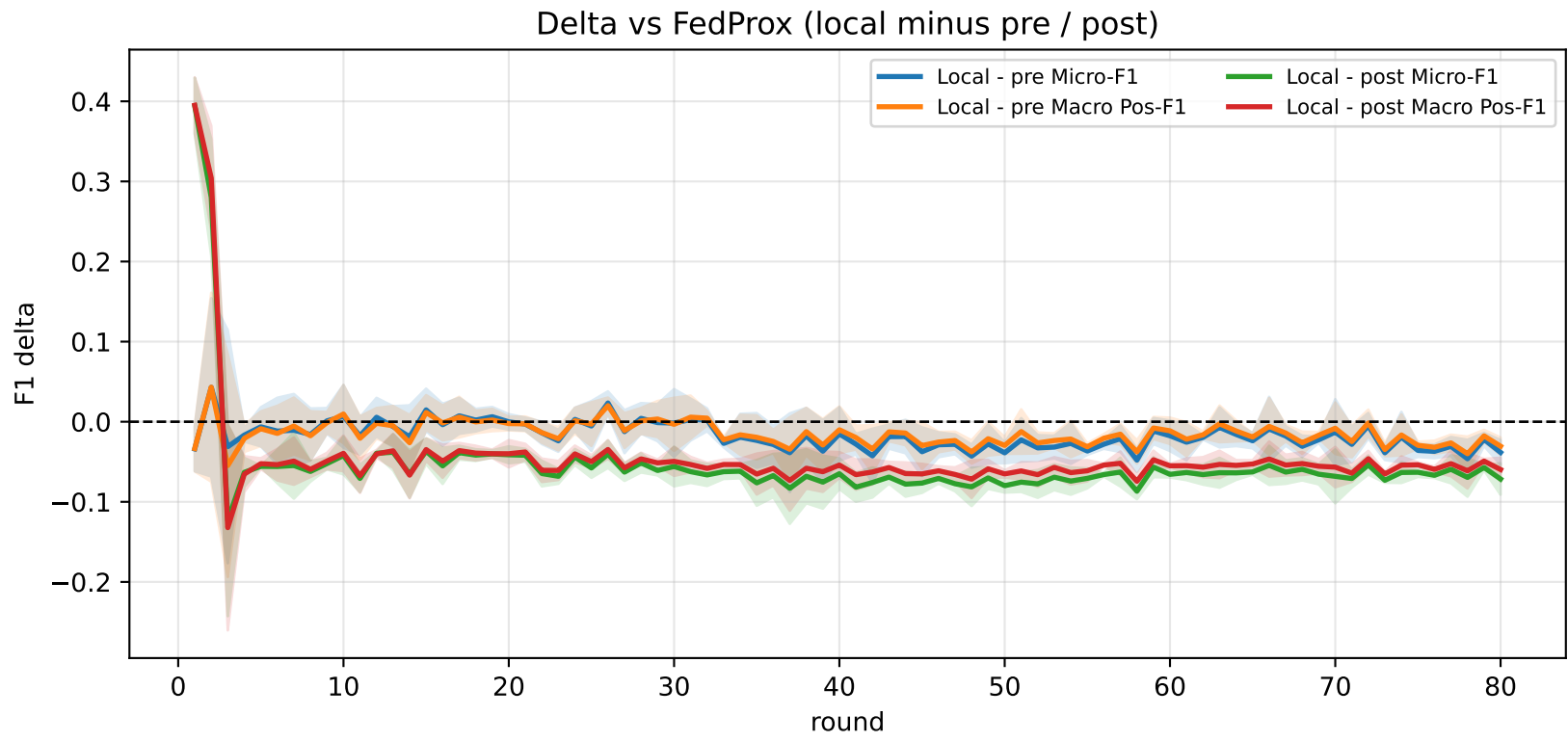
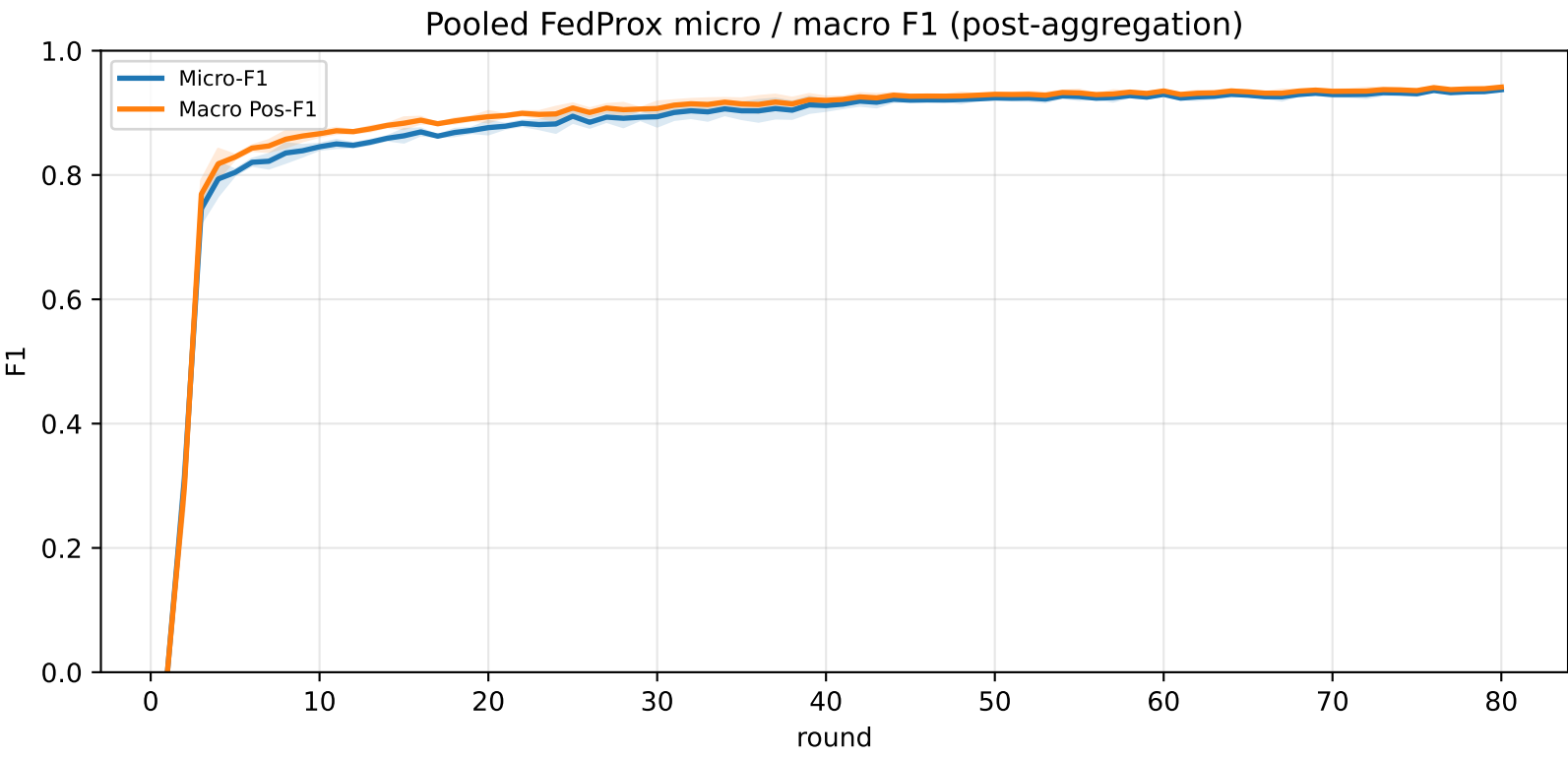
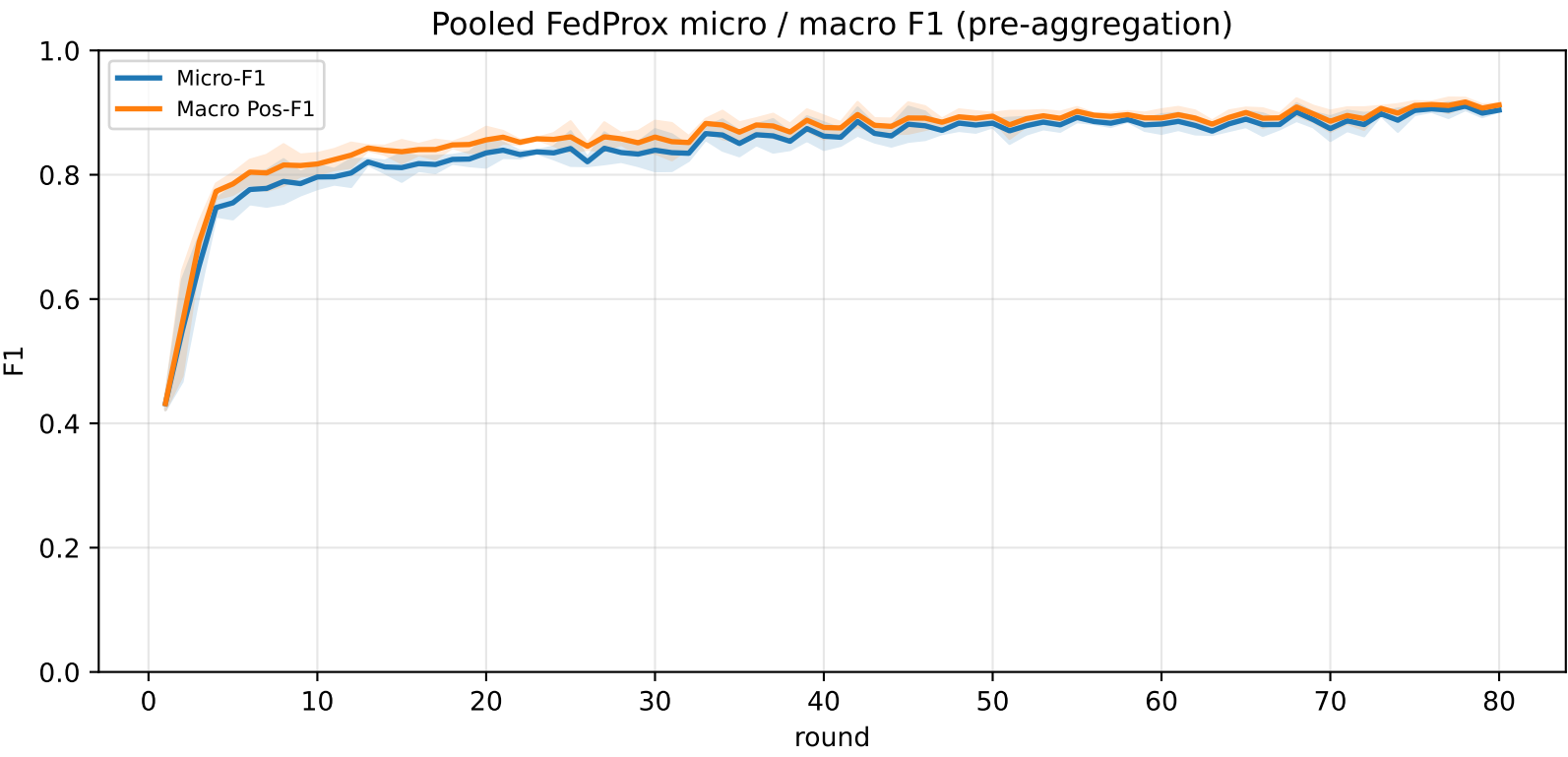
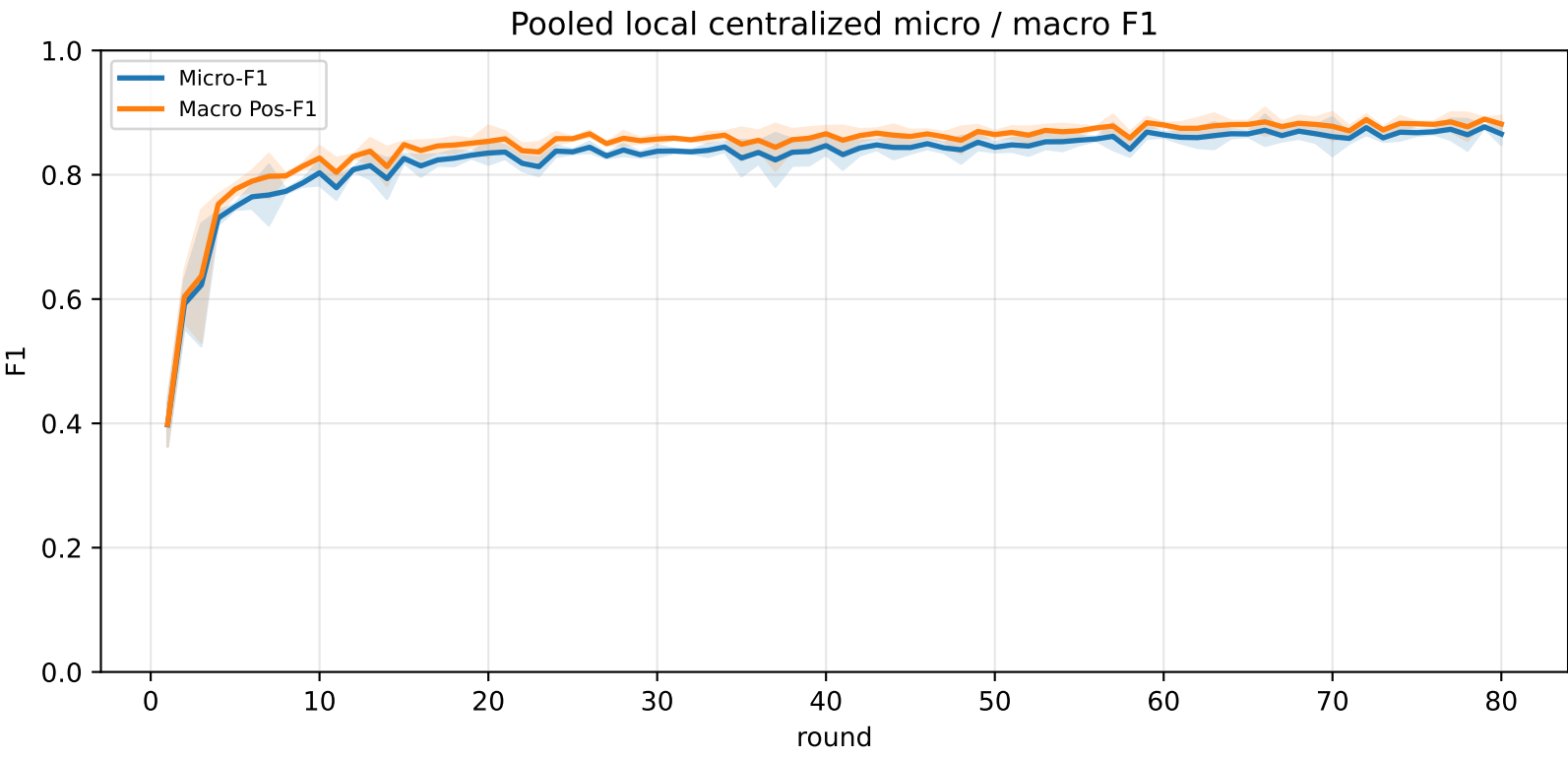
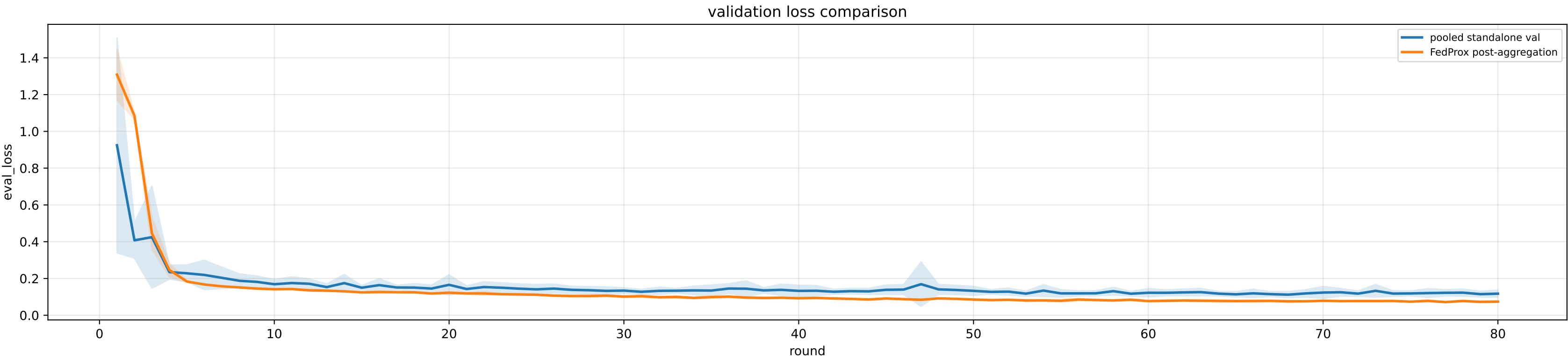
Delta vs FedProx (local minus pre / post)



FedProx diagnostics | family=flat_balanced | n_clients=4 |
model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64

family=flat_balanced | cycle2=8.6%±0.7% | cycle3=10.2%±0.8% | cycle4=9.7%±1.5% | cycle5=9.3%±1.9% | cycle6=8.6%±0.7%

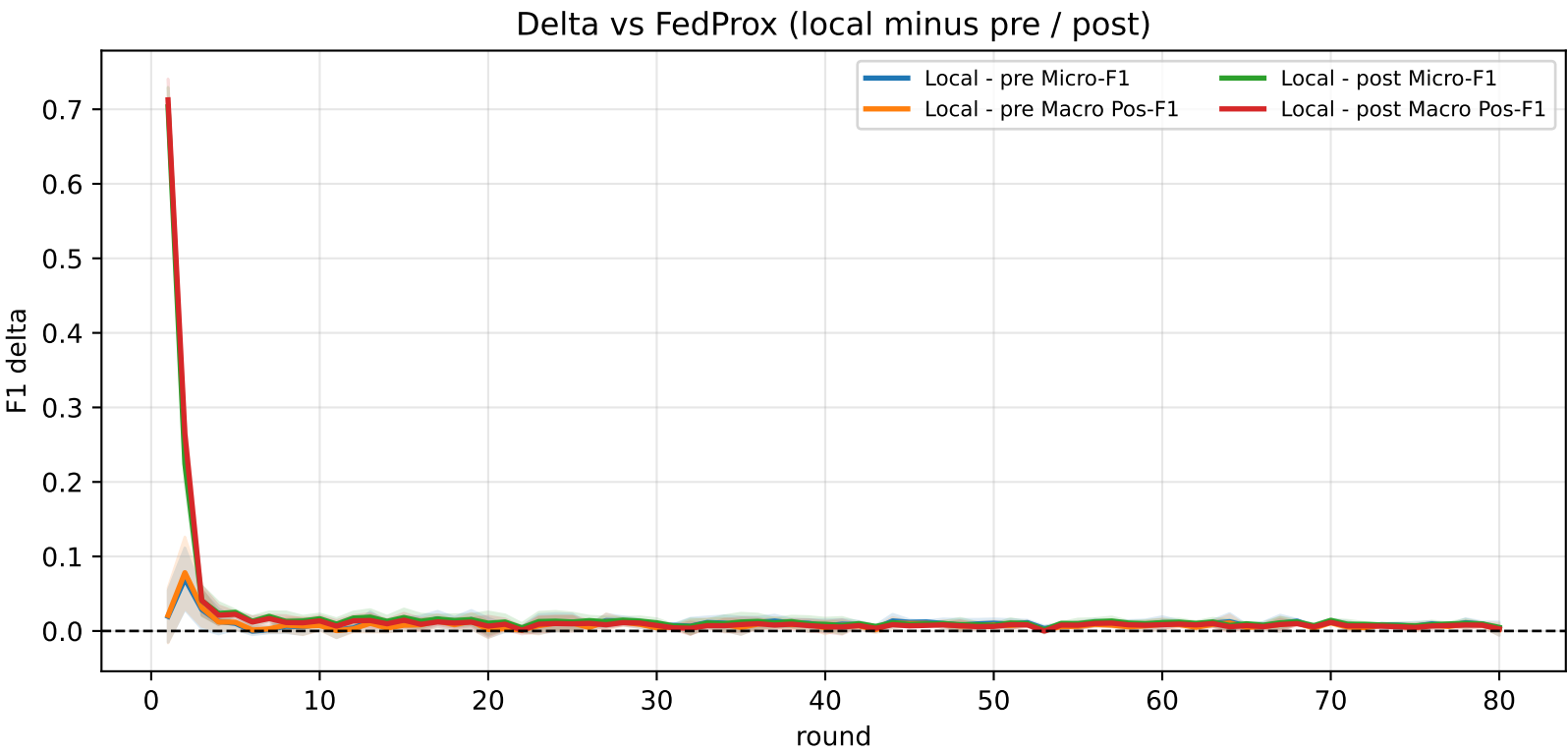
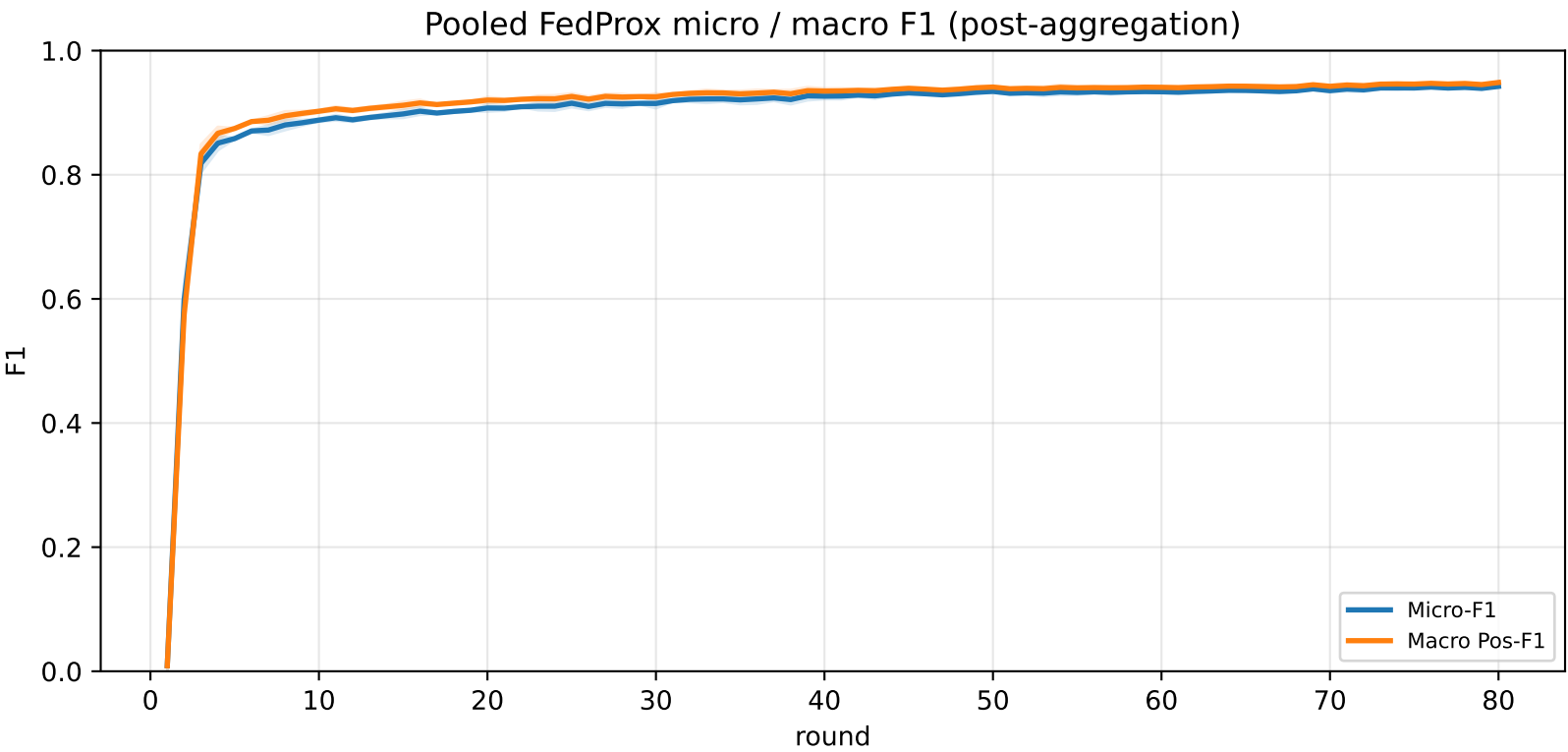
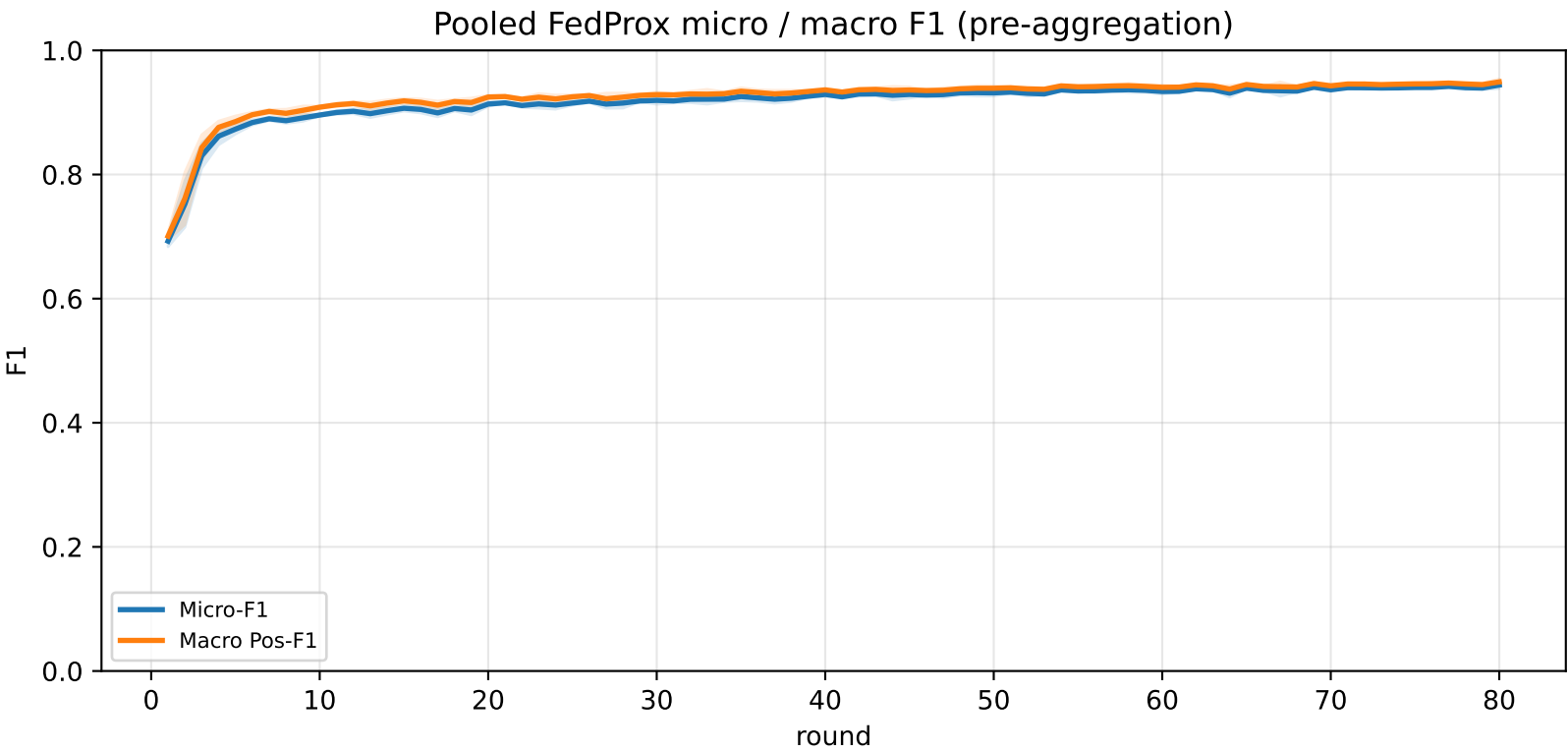
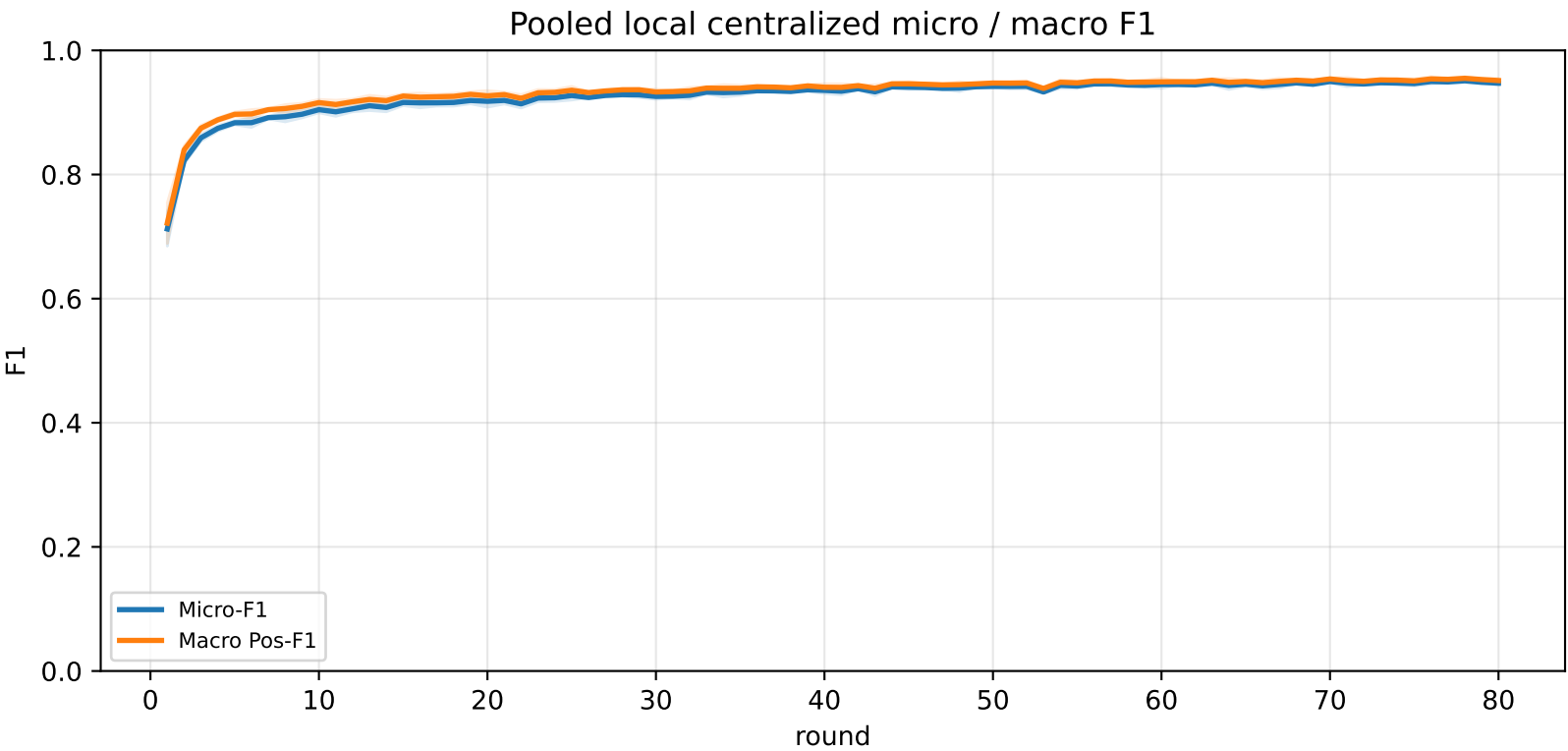
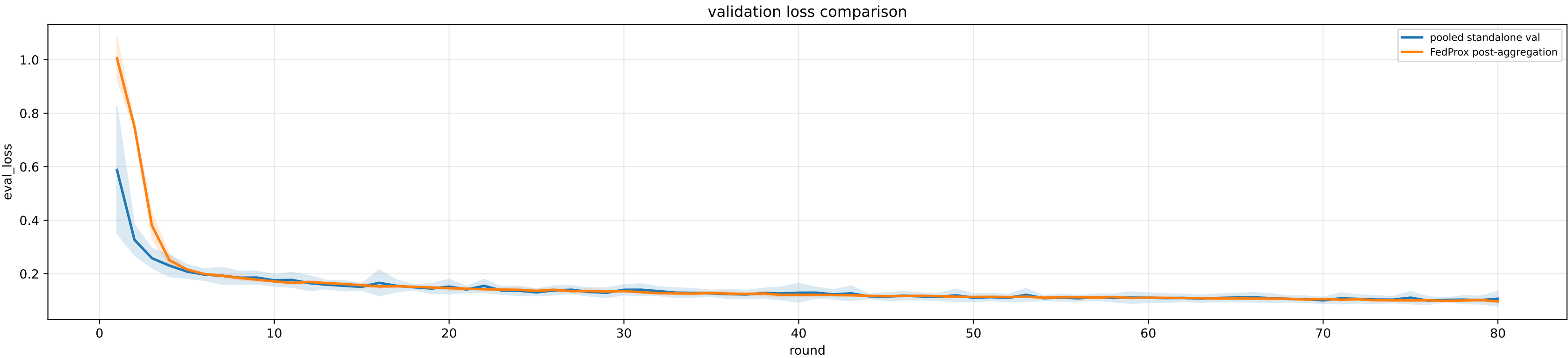
Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	87.6% ± 0.2%	88.9% ± 0.1%	98.6% ± 0.2%	97.0% ± 0.1%	92.6% ± 0.7%	87.0% ± 0.8%	69.2% ± 0.5%
FedProx	93.2% ± 0.1%	93.6% ± 0.1%	99.0% ± 0.1%	98.4% ± 0.4%	96.1% ± 0.6%	93.0% ± 0.2%	81.6% ± 0.3%



FedProx diagnostics | family=mild_increasing | n_clients=4 |
model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64

family=mild_increasing | cycle2=20.6%±0.6% | cycle3=24.0%±0.4% | cycle4=26.6%±0.7% | cycle5=29.1%±0.4% | cycle6=28.9%±0.9%

Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	94.9% ± 0.2%	95.3% ± 0.2%	99.0% ± 0.1%	98.4% ± 0.1%	97.0% ± 0.3%	93.8% ± 0.2%	88.4% ± 0.7%
FedProx	94.1% ± 0.1%	94.6% ± 0.1%	99.0% ± 0.1%	98.4% ± 0.1%	96.0% ± 0.4%	93.4% ± 0.2%	86.4% ± 0.3%



FedProx diagnostics | family=strong_increasing | n_clients=4 |
model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64

family=strong_increasing | cycle2=13.6%±0.4% | cycle3=18.0%±1.5% | cycle4=21.8%±0.4% | cycle5=26.1%±0.7% | cycle6=29.3%±0.4%

Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	93.6% ± 0.2%	94.6% ± 0.1%	98.4% ± 0.2%	97.7% ± 0.1%	95.6% ± 0.1%	92.1% ± 0.6%	89.0% ± 0.4%
FedProx	92.6% ± 0.2%	93.7% ± 0.2%	98.6% ± 0.1%	97.0% ± 0.5%	93.8% ± 0.6%	91.7% ± 0.4%	87.5% ± 0.4%

